# Arithmetic Occupation Spectra and the First Octonionic Nonassociativity Threshold

EABC Research Note

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**Abstract**

We introduce a canonical occupation-space representation of natural numbers based solely on unique prime factorization. For

$$N = \prod_{p \in \mathbb{P}} p^{\nu_p},$$

the associated factor word is

$$W(N) = \sum_{p} \nu_p e_p,$$

with arithmetic occupation numbers

$$\Omega(N) = \sum_{p} \nu_p, \qquad \omega(N) = \#\{p : \nu_p > 0\}.$$

The logarithmic diagonal operator has exact spectral value

$$E_{\times}(N) = \sum_{p} \nu_p \log p = \log N.$$

Prime powers are therefore repeated occupations of one atomic prime mode, while products of distinct primes are genuine multimode states. We prove that in the sector $\Omega = 3$ the three occupation types $p^3$, $p^2q$, and $pqr$ form an exact hierarchy: by alternativity and Artin's theorem, genuine three-generator octonionic nonassociativity is impossible for the first two types and becomes possible for the first time in the squarefree type $pqr$. We then separate the Euclidean tetrahedron volume from the $G_2$ calibration and the octonionic associator defect,

$$V_{\mathrm{tet}} = \frac{1}{6}\sqrt{\det G}, \qquad \varphi(u,v,w) = \langle uv, w\rangle, \qquad \mathfrak{t} = \frac{1}{4}\|[u,v,w]\|^2 = \det G - \varphi^2.$$

No dynamical prime-orbit realization or physical energy interpretation is assumed.

## 1 Factor-word normal form

Let

$$\mathfrak{M} = \bigoplus_{p \in \mathbb{P}} \mathbb{N}_0 e_p$$

be the free commutative monoid on the prime labels $e_p$. By unique prime factorization every $N \geq 1$ has a unique representation

$$N = \prod_{p \in \mathbb{P}} p^{\nu_p}$$

with finitely many nonzero $\nu_p$, and hence a canonical factor word

$$W(N) = \sum_p \nu_p e_p. \tag{1}$$

Composite partial products create no new generators. Thus

$$W(pq) = e_p + e_q, \qquad W(p^2 q) = 2e_p + e_q, \qquad W(pqr) = e_p + e_q + e_r.$$

**Definition 1.1** (Arithmetic occupation numbers)**.** Define

$$\Omega(N) = \sum_p \nu_p, \qquad \omega(N) = \#\{p : \nu_p > 0\}.$$

Thus $\Omega$ counts prime factors with multiplicity, while $\omega$ counts distinct occupied prime modes.

## 2 Logarithmic occupation spectrum

Consider the Hilbert basis $\{|N\rangle : N \in \mathbb{N}\}$ indexed by factor words. Define the diagonal operator

$$H_\times |N\rangle = \left( \sum_p \nu_p \log p \right) |N\rangle. \tag{2}$$

**Proposition 2.1** (Exact arithmetic energy)**.** *For every $N \in \mathbb{N}$,*

$$E_\times(N) = \log N.$$

*Moreover,*

$$E_\times(MN) = E_\times(M) + E_\times(N)$$

*for all $M, N \in \mathbb{N}$.*

*Proof.* Immediate from

$$\log \left( \prod_p p^{\nu_p} \right) = \sum_p \nu_p \log p$$

and $\log(MN) = \log M + \log N$. □

The atomic prime-mode energies are therefore

$$\varepsilon_p = \log p,$$

and a general occupation state has the free additive spectral value

$$E_\times(N) = \sum_p \nu_p \varepsilon_p.$$

This is an arithmetic spectral statement, not yet a physical energy claim.

## 3 Prime, semiprime, and triple occupation sectors

For $\Omega = 1$ one has only prime states

$$p: \quad W(p) = e_p, \qquad E_\times(p) = \log p.$$

For $\Omega = 2$ there are exactly two occupation types:

$$p^2: \quad W = 2e_p, \quad \omega = 1, \quad E_\times = 2\log p,$$

$$pq\ (p \neq q): \quad W = e_p + e_q, \quad \omega = 2, \quad E_\times = \log p + \log q.$$

Thus $p^2$ is repeated occupation of one mode, whereas $pq$ is a genuine two-mode state.

For $\Omega = 3$ unique factorization yields exactly the partition types 3, $2+1$, and $1+1+1$:

| Arithmetic type | Factor word | $\omega(N)$ | $E_\times(N)$ |
|---|---|---|---|
| $p^3$ | $3e_p$ | 1 | $3\log p$ |
| $p^2q$, $p \neq q$ | $2e_p + e_q$ | 2 | $2\log p + \log q$ |
| $pqr$, pairwise distinct | $e_p + e_q + e_r$ | 3 | $\log p + \log q + \log r$ |

If one uses bosonic occupation notation, these correspond formally to

$$|3_p\rangle, \qquad |2_p, 1_q\rangle, \qquad |1_p, 1_q, 1_r\rangle$$

in $\mathrm{Sym}^3 \mathcal{H}_1$. This symmetric occupation structure must not be identified with exterior powers appearing in Fredholm determinant expansions.

## 4 Octonionic realization of occupied modes

Assume that each distinct occupied prime mode is assigned a fixed imaginary octonionic direction

$$p \longmapsto u_p \in \mathrm{Im}\,\mathbb{O}.$$

No canonicity of this assignment is assumed.

For $x, y, z \in \mathbb{O}$, define the associator

$$[x, y, z] = (xy)z - x(yz).$$

The octonions are alternative, so

$$[x, x, y] = [y, x, x] = 0.$$

Moreover, Artin's theorem implies that every subalgebra generated by two elements of an alternative algebra is associative.

**Theorem 4.1** (Occupation–associativity threshold)**.** *Let $N$ satisfy $\Omega(N) = 3$, and assign an octonionic generator to every distinct occupied prime mode. Then:*

*(i) for $N = p^3$, every local three-word associator vanishes;*

*(ii) for $N = p^2q$ with $p \neq q$, every associator built from the two occupied generators $u_p, u_q$ vanishes;*

*(iii) for $N = pqr$ with $p, q, r$ pairwise distinct, a nonzero associator $[u_p, u_q, u_r]$ becomes possible.*

*Consequently,*

$$\omega(N) \le 2 \quad \Longrightarrow \quad \textit{genuine three-generator nonassociativity is impossible,}$$

*whereas* $\omega(N) = 3$ *is the first occupation threshold at which it may occur.*

*Proof.* For $p^3$, all three entries are the same generator, hence alternativity gives $[u_p, u_p, u_p] = 0$. For $p^2q$, at most the two generators $u_p, u_q$ occur. By Artin's theorem the subalgebra generated by them is associative, so every three-word associator formed from these generators vanishes. For $pqr$, three distinct generators occur and Artin's two-generator theorem no longer forces associativity. Thus nonzero associators are possible, though not necessary. □

*Remark* 4.2. The squarefree condition does not imply nonassociativity. Three distinct octonionic generators can still lie in a common associative quaternionic subalgebra. The theorem identifies the first *possible* nonassociative occupation type, not a sufficient arithmetical criterion.

## 5 $G_2$ geometry of the three-mode configuration

Let $u, v, w \in \operatorname{Im}\mathbb{O} \cong \mathbb{R}^7$. Write

$$G = (\langle u_i, u_j \rangle)_{i,j=1}^{3}$$

for their Gram matrix. The Euclidean volume of the simplex $\{0, u, v, w\}$ is

$$V_{\text{tet}}(u, v, w) = \frac{1}{6}\sqrt{\det G}. \tag{3}$$

This is a purely metric invariant and must not be confused with the associator defect.

The standard associative $G_2$ three-form is, up to orientation convention,

$$\varphi(u, v, w) = \langle uv, w \rangle. \tag{4}$$

Define the nonassociativity defect

$$\mathfrak{t}(u, v, w) = \frac{1}{4}\|[u, v, w]\|^2. \tag{5}$$

For imaginary octonions one has the identity

$$\boxed{\mathfrak{t} = \det G - \varphi^2.} \tag{6}$$

Equivalently,

$$\boxed{\mathfrak{t} = 36V_{\text{tet}}^2 - \varphi^2.} \tag{7}$$

Thus $V_{\text{tet}}$, $\varphi$, and $\mathfrak{t}$ are related observables, not three independent invariants.

For an orthonormal triple, $G = I_3$ and hence

$$V_{\text{tet}} = \frac{1}{6}, \qquad \mathfrak{t} = 1 - \varphi^2, \qquad 0 \le \mathfrak{t} \le 1.$$

General orthonormal $G_2$ triples need not be binary: intermediate values of $|\varphi|$ and $\mathfrak{t}$ are possible.

## 6 Discrete Fano basis sector

Let $e_1, \ldots, e_7$ be the seven imaginary basis units in a fixed octonion multiplication table. There are

$$\binom{7}{3} = 35$$

unordered triples of distinct basis directions. Exactly seven are Fano-line triples and twenty-eight are non-Fano triples.

Every basis triple is orthonormal, hence all thirty-five associated Euclidean tetrahedra satisfy

$$V_{\text{tet}} = \frac{1}{6}.$$

The $G_2$ multiplication nevertheless separates them:

| Type | Number | $\det G$ | $\lvert\varphi\rvert$ | $\mathfrak{t}$ |
|---|---|---|---|---|
| Fano line | 7 | 1 | 1 | 0 |
| Non-Fano basis triple | 28 | 1 | 0 | 1 |

Thus the thirty-five basis tetrahedra are Euclidean-volume equivalent but octonionically inequivalent. The distinction lies in the calibrated multiplication structure, not in ordinary metric volume.

## 7 Configuration invariants versus arithmetic invariants

For a fixed triple $(u_p, u_q, u_r)$ the quantities

$$G, \qquad V_{\text{tet}}, \qquad \varphi, \qquad \mathfrak{t}$$

are mathematically well-defined. They are therefore configuration invariants.

They are not yet invariants of the bare prime triple $(p, q, r)$. To obtain such an arithmetic invariant one would need a canonical or quotient-stable assignment

$$p \longmapsto u_p$$

or an invariant averaging procedure over a mathematically specified representation fiber. A choice of one four-square representation

$$p = a^2 + b^2 + c^2 + d^2$$

and an arbitrary embedding of those coordinates into $\operatorname{Im}\mathbb{O}$ does not supply such canonicity.

In particular, quotienting a single unit vector by the full group $G_2 = \operatorname{Aut}(\mathbb{O})$ does not solve the problem: $G_2$ acts transitively on the unit sphere $S^6 \subset \operatorname{Im}\mathbb{O}$. Relevant $G_2$ invariants arise only for multi-vector configurations.

## 8 Prime powers, multimode states, and possible orbit realizations

The exact arithmetic identity

$$E_\times(p^k) = k \log p$$

exhibits prime powers as the occupation ladder

$$|1_p\rangle,\ |2_p\rangle,\ |3_p\rangle, \ldots$$

along one prime mode. Mixed products activate several modes.

The formula resembles the repetition law

$$\ell(\gamma^k) = k\ell(\gamma)$$

for periodic orbits. However, the existence of a dynamics with

$$p \mapsto \gamma_p, \qquad \ell(\gamma_p) = \log p$$

is a separate realization problem and is not assumed here. In particular, the arithmetic equality $E_\times(p) = \log p$ must not be identified with a physically or dynamically realized primitive period without proof.

Likewise, bosonic occupation sectors

$$\mathrm{Sym}^k \mathcal{H}_1$$

must not be conflated with Fredholm exterior powers

$$\Lambda^k \mathcal{H}.$$

Any bridge between these constructions requires an independent theorem.

## 9 Epistemic status and open gates

The following statements are established within the algebraic framework:

- the canonical factor word $W(N) = \sum_p \nu_p e_p$;
- the arithmetic occupation data $\Omega(N)$ and $\omega(N)$;
- the exact logarithmic spectrum $E_\times(N) = \log N$;
- the complete $\Omega = 3$ decomposition into $p^3$, $p^2q$, and $pqr$;
- Theorem 4.1;
- for a fixed octonionic triple, the geometric quantities $G$, $V_{\mathrm{tet}}$, $\varphi$, and $\mathfrak{t}$ and identity (6);
- the discrete $35 = 7 + 28$ Fano-basis classification.

The following remain open realization problems:

1. a canonical arithmetic assignment $p \mapsto u_p$;
2. representation-independence of the octonionic configuration invariants;
3. a periodic-orbit realization with primitive periods $\log p$;
4. a physical interpretation of $E_\times$ as measured energy;
5. any identification between factor-word Fock occupation and Fredholm exterior-power combinatorics.

## 10 Conclusion

Unique prime factorization determines a canonical occupation vector, and logarithmic evaluation converts multiplicative arithmetic into an exactly additive spectrum:

$$\boxed{N = \prod_p p^{\nu_p} \quad \longmapsto \quad W(N) = \sum_p \nu_p e_p, \qquad E_\times(N) = \sum_p \nu_p \log p = \log N.}$$

Prime powers encode repeated occupation of one atomic mode, whereas mixed products encode genuine mode multiplicity. At occupation number three, alternativity and Artin's theorem force the $p^3$ and $p^2q$ sectors to remain associative and identify the squarefree $pqr$ sector as the first one in which a genuine octonionic associator can occur.

The resulting structural principle is

$$\boxed{\text{arithmetic mode complexity} \implies \text{first possible algebraic nonassociativity threshold.}}$$

This conclusion is entirely algebraic. The remaining task is the arithmetic anchoring of the octonionic configuration, not the internal $G_2$ geometry itself.